

Experiment 'p_6_9_arrays_equal_seq' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_9_arrays_equal_seq

Natural language statement: Write a method that checks whether two arrays have the same elements in the same order.

Method signature: p_6_9_arrays_equal_seq(first: seq<int>, second: seq<int>) returns (are_equal: bool)

Ensures

- are_equal == (|first| == |second| && forall i :: 0 <= i < |first| ==> first[i] == second[i])

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that checks whether two arrays have the same elements in the same order.

The signature should be:

`p_6_9_arrays_equal_seq(first: seq<int>, second: seq<int>) returns (are_equal: bool)`

The method should respect the following contract:

`ensures are_equal == (|first| == |second| && forall i :: 0 <= i < |first| ==> first[i] == second[i])`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dnli2j0Fb2STEWtKTY9CzGzIf1a19

Model: gpt-5.5-2026-04-23

Created at: 1780577014

Finish reason: stop

Usage: 326 tokens in, and 1853 tokens out

```
{
  if |first| != |second| {
    are_equal := false;
    return;
  }

  are_equal := true;
  var i := 0;
  while i < |first|
    invariant |first| == |second|
    invariant 0 <= i <= |first|
    invariant are_equal
    invariant forall j :: 0 <= j < i ==> first[j] == second[j]
  {
    if first[i] != second[i] {
      are_equal := false;
      assert 0 <= i < |first|;
      assert !(forall j :: 0 <= j < |first| ==> first[j] == second[j]);
      return;
    }
  }
}
```

```

    assert forall j :: 0 <= j < i + 1 ==> first[j] == second[j] by {
      forall j | 0 <= j < i + 1
        ensures first[j] == second[j]
      {
        if j < i {
        } else {
          assert j == i;
        }
      }
    }
    i := i + 1;
  }

  assert i == |first|;
  assert forall j :: 0 <= j < |first| ==> first[j] == second[j];
}

```

Final Program

```

method p_6_9_arrays_equal_seq(first: seq<int>, second: seq<int>) returns (are_equal: bool
  ↪ )
  ensures are_equal == (|first| == |second| && forall i :: 0 <= i < |first| ==>
    ↪ first[i] == second[i])
{
  if |first| != |second| {
    are_equal := false;
    return;
  }

  are_equal := true;
  var i := 0;
  while i < |first|
    invariant |first| == |second|
    invariant 0 <= i <= |first|
    invariant are_equal
    invariant forall j :: 0 <= j < i ==> first[j] == second[j]
  {
    if first[i] != second[i] {
      are_equal := false;
      assert 0 <= i < |first|;
      assert !(forall j :: 0 <= j < |first| ==> first[j] == second[j]);
      return;
    }

    assert forall j :: 0 <= j < i + 1 ==> first[j] == second[j] by {
      forall j | 0 <= j < i + 1
        ensures first[j] == second[j]
      {
        if j < i {

```

```
        } else {
            assert j == i;
        }
    }
    i := i + 1;
}

assert i == |first|;
assert forall j :: 0 <= j < |first| ==> first[j] == second[j];
}
```

Total Token Usage

Input tokens: 326

Output tokens: 1853

Reasoning tokens: 1535

Sum of 'total tokens': 2179

Experiment Timings

Overall Experiment started at 1780577013367, ended at 1780577080412, lasting 67045ms (67.05 seconds)

Iteration #1 started at 1780577013369, ended at 1780577080410, lasting 67041ms (67.04 seconds)